



ADVANCED MALWARE ANALYSIS AND INCIDENT RESPONSE EVALUATION USING ANY.RUN

**SOC ANALYST TASK 4
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ABSTRACT

This report details the advanced malware analysis exercise focusing on both static and dynamic analysis of a malware sample within a controlled virtual environment. Utilizing sandbox tools such as ANY.RUN and a suite of forensic tools including Process Explorer, Regshot, and FakeNet-NG, the analysis was conducted to observe and document the malware's behavior. The static analysis provided insights into the file's metadata and code structure, while dynamic analysis revealed real-time changes to the file system, network activity, and registry modifications. A hybrid approach integrated both analysis types to enhance understanding of the malware's impact. Comparative analysis was conducted by contrasting findings with existing reports from reputable sources to verify accuracy and identify unique observations. The comprehensive report includes a detailed incident response plan outlining detection, containment, eradication, and recovery strategies. The results underscore the effectiveness of the analysis techniques and offer a structured framework for addressing similar threats in future scenarios.

STEPS

1. Preparing the Virtual Environment and Sandbox

Virtual Environment:

A Windows 10 virtual machine (VM) was set up using VirtualBox to ensure that the malware execution was isolated from the host system.

Sandbox Tool:

ANY.RUN: Used for dynamic analysis, ANY.RUN provided a controlled environment to observe real-time malware execution. The tool's capabilities included monitoring process creation, file system changes, network traffic, and registry modifications. It also allowed for interactive analysis and detailed behavioral tracking.

2. Malware Download and Preparation

Selecting the Malware Sample: A RAT (Remote Access Trojan), RemcosRAT to be specific, sample was downloaded from [MalwareBazaar](#), identified by hash as 66c50343775c162862ac27a735c66927a9b3fda4a05cd0eaa21fecbca3f6c490. This sample was chosen for its known malicious behavior and relevance to current threats. Malware of the RAT kind called Remcos is used by attackers to carry out remote tasks on compromised systems. Updates for this malware are released nearly every month, demonstrating how continuously it is kept up to date.

Preparing the Malware File: The malware was compressed into a ZIP file and password-protected to prevent accidental execution. The ZIP file was named

66c50343775c162862ac27a735c66927a9b3fda4a05cd0eaa21fecbca3f6c490.zip with the password "infected". The source URL for the malware is: [malware](#).

3. Tools Used

- **Process Explorer:** Monitored active processes and their attributes during malware execution. Key observations included new process creation and resource usage.
- **Regshot:** Tracked registry changes before and after the malware execution, capturing modifications that include new registry keys and values related to persistence.
- **Process Monitor:** Captured real-time file system, registry, and process/thread activity, highlighting file modifications, network connections, and process creation.
- **FakeNet-NG:** Simulated network traffic to capture any network interactions initiated by the malware, including outbound connections to command-and-control servers.
- **Netstat:** Used to inspect network connections made by the malware, identifying any unusual or malicious network activity.
- **Dumplt:** Captured memory dumps for forensic analysis, allowing for the examination of in-memory artifacts and potential malicious code.

4. Static Analysis

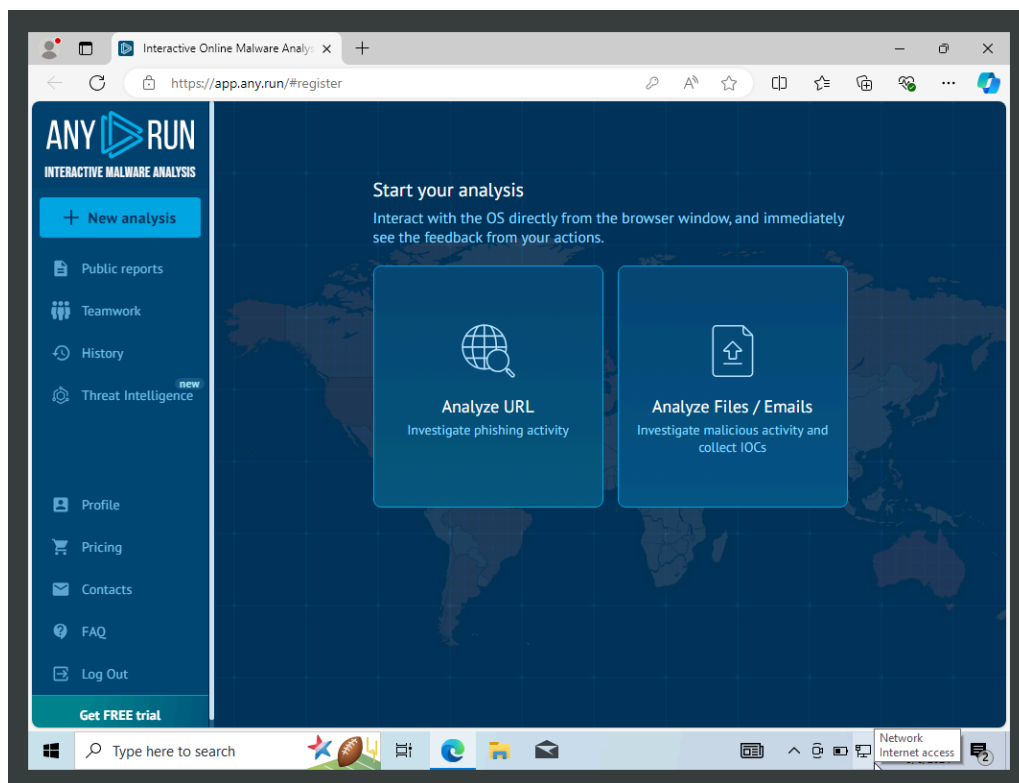
- **Examining File Metadata:** The static analysis of the malware sample was performed using ExifTool, which provided detailed metadata about the file. Key findings include the following:
 - **File Type:** The file is identified as a Win32 executable with a MIME type of application/octet-stream.
 - **File Size:** The file size is 852 KiB, suggesting a relatively small executable.
 - **Time Stamp:** The file contains a suspicious future timestamp of 2087:10:31, which indicates tampering or attempts to evade detection by anti-malware tools relying on time-based heuristics.
 - **PE Format:** The file uses the PE32 format, indicating a 32-bit executable.
 - **Machine Type:** The file is compatible with Intel 386 or later machines.
 - **Subsystem:** The executable is designed to run within a Windows GUI environment.
 - **File Version Information:** The file is labeled as version 1.0.0.0, and the internal and original names are Grator.exe. The lack of company name and comments fields is typical of malicious files trying to mask their origin.
 - **Copyright Information:** The file includes a copyright mark of "© 2024", suggesting an attempt to appear legitimate, although this information is likely forged.

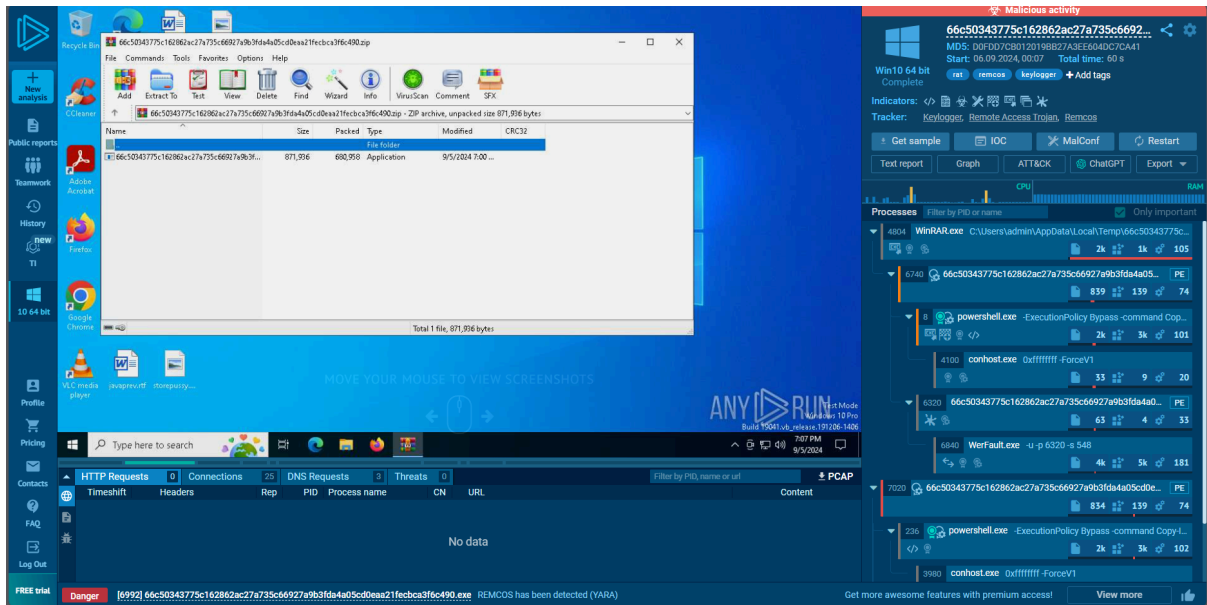
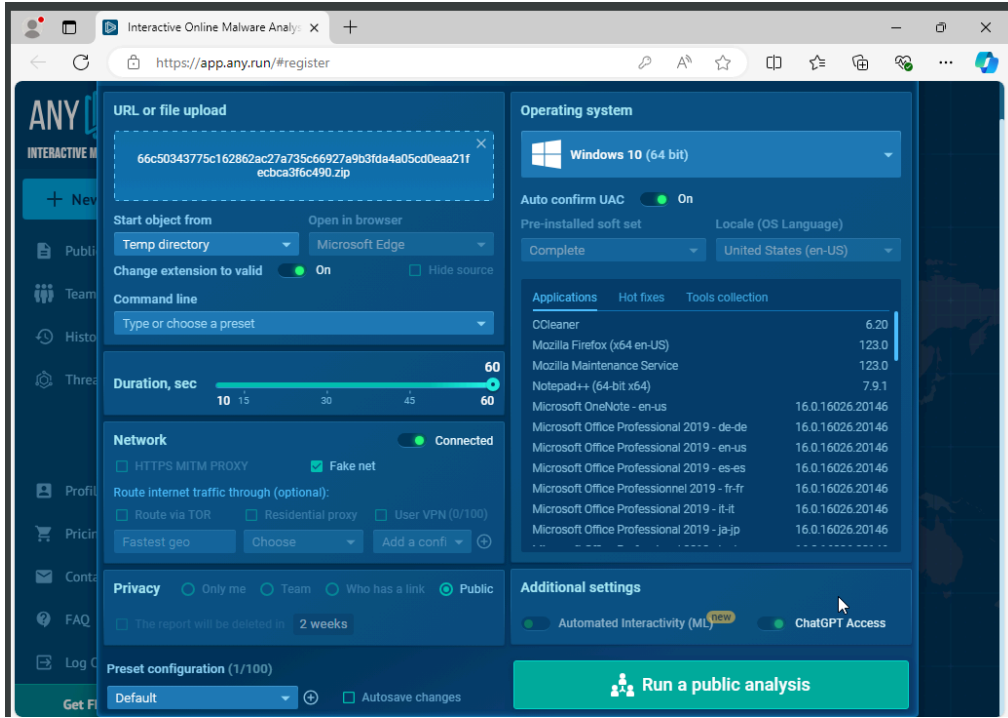
These findings indicate that the malware is a well-crafted executable designed to execute on 32-bit Windows systems. The irregular timestamp and lack of legitimate metadata hint at potential malicious intent. Further dynamic analysis is required to confirm its behavior.

- **Disassembling the Malware:** IDA was used to disassemble the file. The disassembled malware is a .NET-based Portable Executable (PE) file, likely developed using Visual C++. It imports functions from ``mscorlib.dll``, indicating reliance on the .NET framework and managed code executed by the Common Language Runtime (CLR). The executable's code section likely includes the main logic and routines for decrypting or unpacking additional malicious components. It uses instructions for the Intel 80386 architecture, with assembly setups like ``model flat``, indicating it is built for modern 32-bit systems. The malware follows typical .NET execution flows, such as environment setup, payload decryption, and possibly modifying system settings for persistence. It may communicate with a remote command-and-control (C2) server, steal sensitive data, and utilize anti-debugging techniques or obfuscation to hinder detection and analysis. This suggests the malware is designed for persistence, data theft, and potential damage within environments that commonly use the .NET framework.

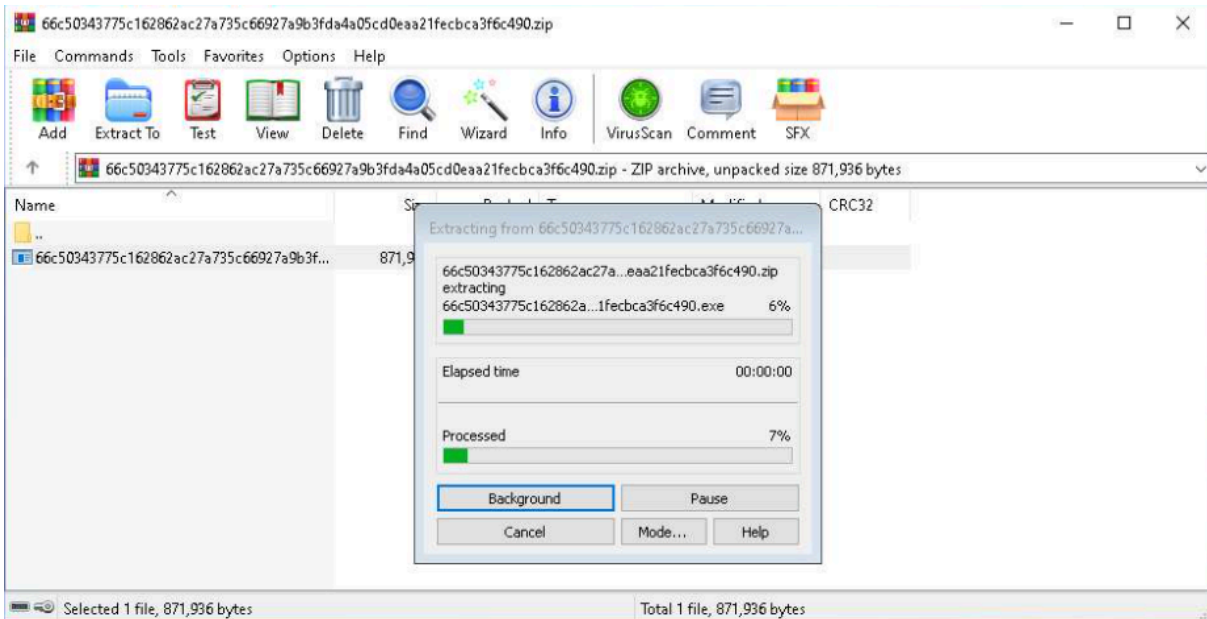
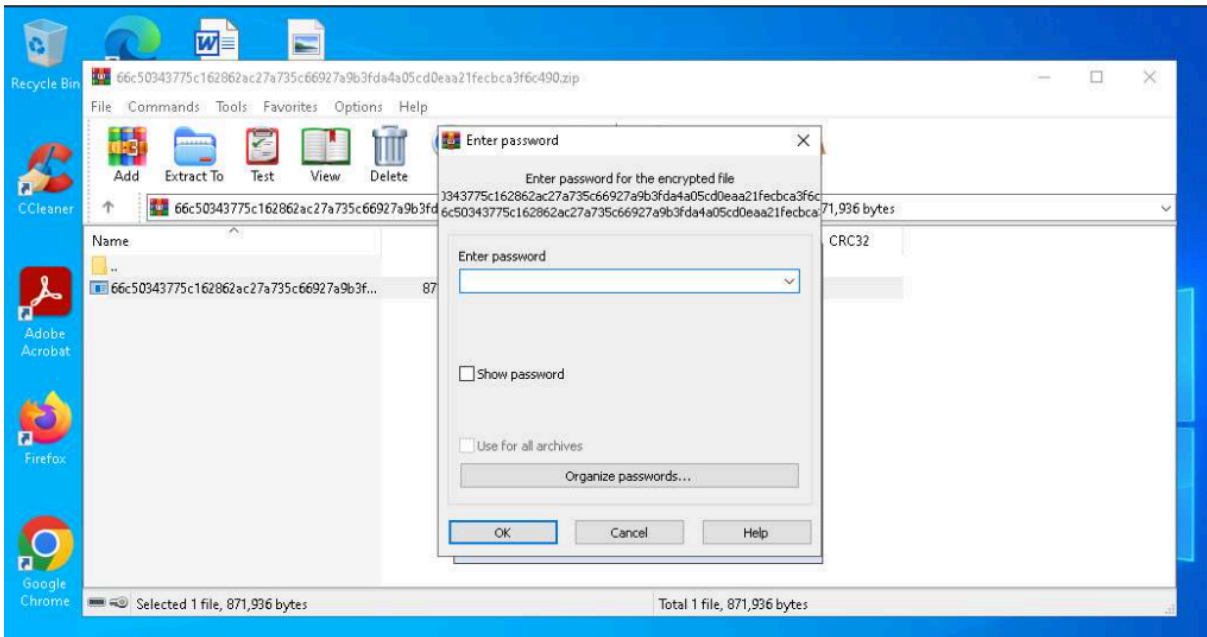
5. Dynamic Analysis

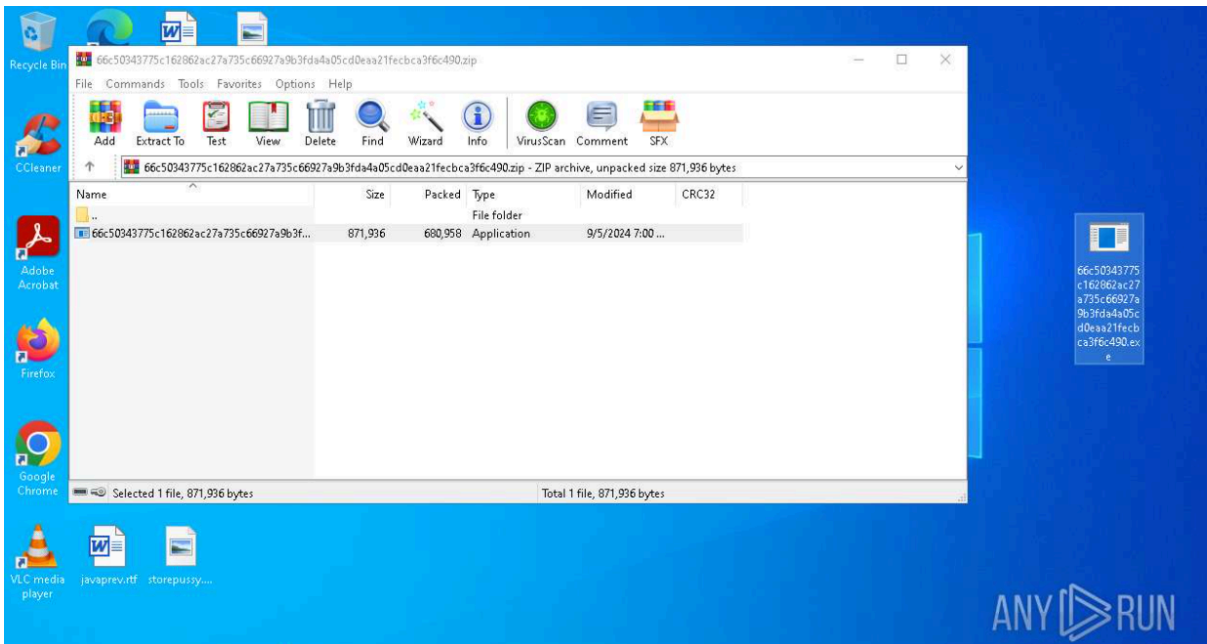
Proceed to [any.run](https://app.any.run) and Create a New Analysis.





Here you can see the real-time behavior of the malware.

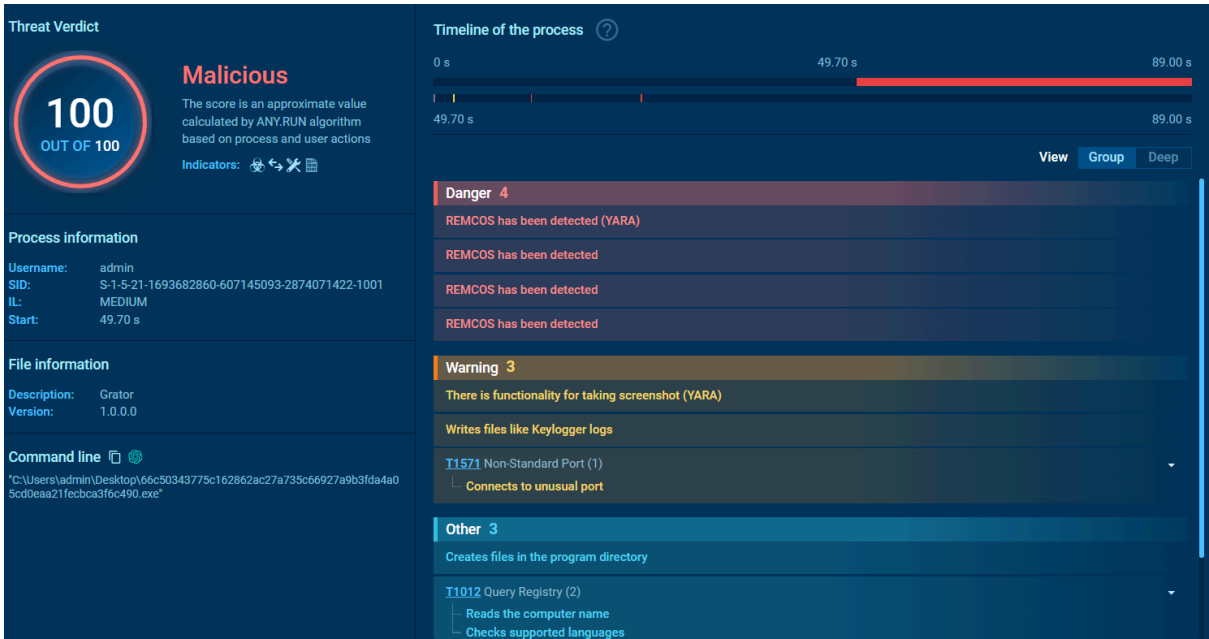




A malicious process 66c50343775c162862... is seen to be connecting to IP 107.175.229.139 on port 8823 via TCP in the connections, possibly communicating with a C2 server.

Filter by PID, domain, name or ip												PCAP
HTTP Requests	0	Connections	25	DNS Requests	3	Threats	0					
Timeshift	Protocol	Rep	PID	Process name	CN	IP	Port	Domain	ASN	Traffic		
8220 ms	TCP	✓	7008	svchost.exe	✓	51.124.78.146	443	settings-win.data.micr...	MICROSOFT-CORP-MS...	309 B	1.43 Kb	
10327 ms	TCP	✓	7008	svchost.exe	✓	51.124.78.146	443	settings-win.data.micr...	MICROSOFT-CORP-MS...	309 B	1.43 Kb	
18435 ms	TCP	Malicious	6840	WerFault.exe	✓	20.42.73.29	443	watson.events.data.mi...	MICROSOFT-CORP-MS...	497 B	1.43 Kb	
45055 ms	TCP	✗	6992	66c50343775c162862...	✗	107.175.229.139	8823	-	AS-COLOCROSSING	No Data		

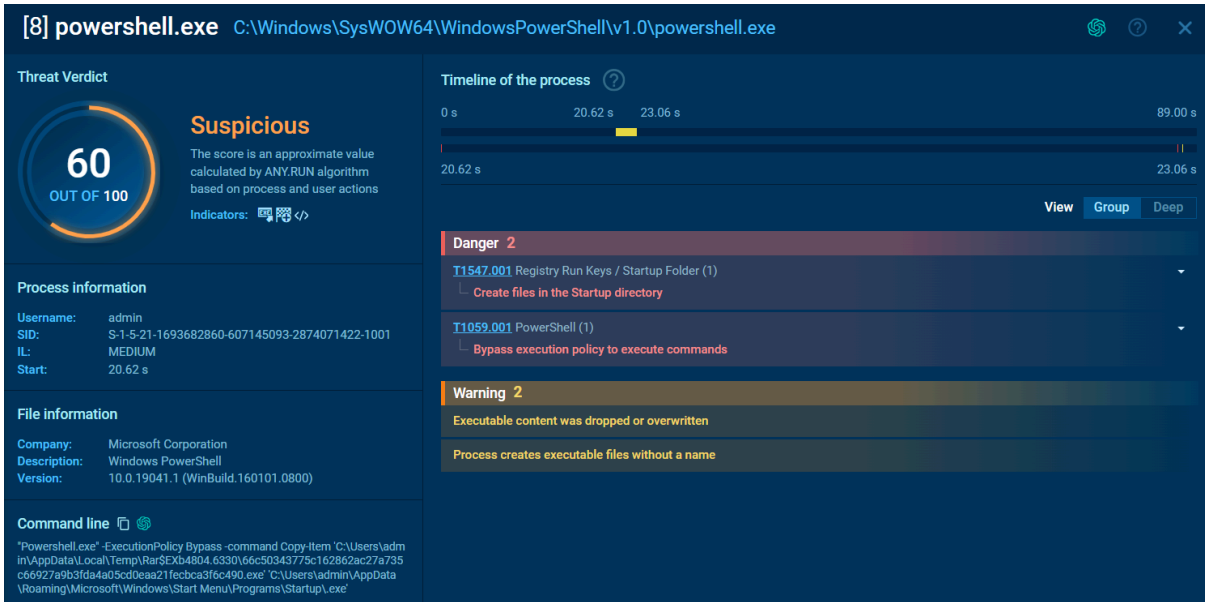
Upon inspecting it further, we get to know that there is a 100/100 malicious score for the process `66c50343775c162862...`, identified as **Remcos RAT**. It takes screenshots, logs keystrokes, and connects to unusual ports. Detected multiple times by YARA, it exhibits dangerous behavior, including creating files and querying the registry, posing a serious security threat.



A malicious process **WinRAR.exe** is seen modifying the file 66c50343775c162862ac27a735c66927a9b3fda4a05cd0eaa21fecbca3f6c490.exe in the directory C:\Users\admin\AppData\Local\Temp\Rar\$EXb4804.6330, indicating potential malware execution. The modified executable file is 851 Kb in size. Similarly, there are more file modifications by this malware too.

Files modification					14	☒ Only important	Filter by filename
NETWORK	Timeshift	PID	Process name	Filename	Content		
	13795 ms	4804	WinRAR.exe	C:\Users\admin\AppData\Local\Temp\Rar\$EXb4804.6330\66c50343775c162862ac27a735c66927a9b3fda4a05cd0eaa21fecbca3f6c490.exe	851 Kb	executable	
FILES	15952 ms	8	powershell.exe	C:\Users\admin\AppData\Local\Temp_PSScriptPolicyTest_bgnzmy4b.pwo.ps1	60 b	text	
	15952 ms	8	powershell.exe	 C:\Users\admin\AppData\Local\Temp_PSScriptPolicyTest_a5t0yxhi.01k.psm1		60 b text	
DEBUG	16858 ms	6840	WerFault.exe	C:\ProgramData\Microsoft\Windows\WER\Temp\WERDE33.tmp.dmp	43.0 Kb	binary	
	16874 ms	6840	WerFault.exe	C:\ProgramData\Microsoft\Windows\WER\Temp\WERDEC0.tmp.WERInternalMetadata.xml	6.30 Kb	xml	
	16921 ms	6840	WerFault.exe	C:\ProgramData\Microsoft\Windows\WER\Temp\WERDEE1.tmp.xml	4.91 Kb	xml	

A suspicious process **powershell.exe** is seen executing a command to copy the file ``66c50343775c162862ac27a735c66927a9b3fda4a05cd0eaa21fecbca3f6c490.exe`` to the Startup directory at ``C:\Users\admin\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup.exe``, bypassing execution policies. This behavior suggests persistence tactics typically associated with malware.



Files modification		14	Only important		Filter by filename	
NETWORK	Timeshift	PID	Process name	Filename	Content	
	17483 ms	8	powershell.exe	C:\Users\admin\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup.exe	851 Kb	executable
FILES	17530 ms	8	powershell.exe	C:\Users\admin\AppData\Local\Microsoft\Windows\PowerShell\StartupProfileData-NonInteractive	17.4 Kb	binary
	17561 ms	6840	WerFault.exe	C:\ProgramData\Microsoft\Windows\WER\ReportQueue\AppCrash_66c50343775c1628_2cadcf84bc54896ca67e63937f658e6af53ec_a542d378_3d060180-ceab-4316-b32c-569c4c6b5994\Report.wer	-	
DEBUG	17686 ms	6840	WerFault.exe	C:\Users\admin\AppData\Local\CrashDumps\66c50343775c162862ac27a735c66927a9b3fda4a05cd0eaa21fecbca3f6c490.exe.6320.dmp	417 Kb	binary
	38593 ms	4804	WinRAR.exe	C:\Users\admin\AppData\Local\Temp\Rar\$DRb4804.8882\66c50343775c162862ac27a735c66927a9b3fda4a05cd0eaa21fecbca3f6c490.exe	851 Kb	executable
43619 ms	236	powershell.exe	C:\Users\admin\AppData\Local\Temp_PSScriptPolicyTest_fln0hju.2uy.ps1	60 b	text	
43619 ms	236	powershell.exe	C:\Users\admin\AppData\Local\Temp_PSScriptPolicyTest_3l5yti24.5ye.psm1	60 b	text	
49229 ms	6992	66c50343775c162862...	C:\ProgramData\remcos\logs.dat	144 b	binary	

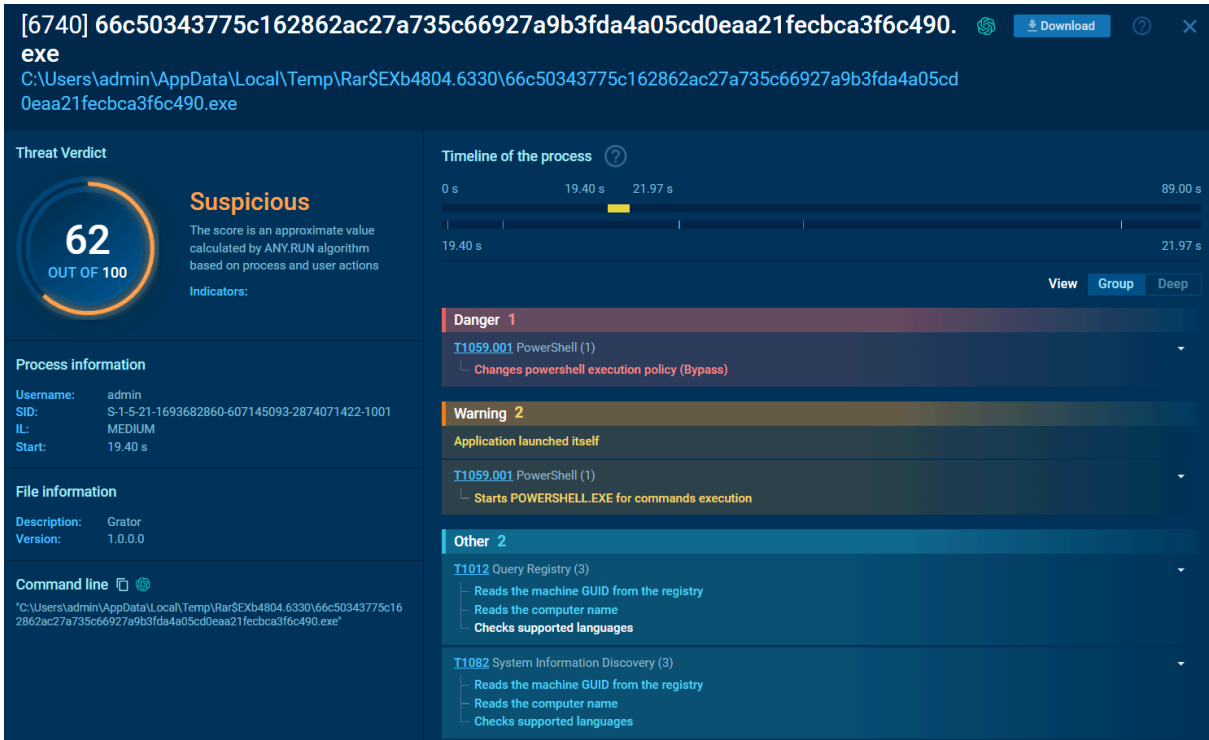
DNS requests were made to multiple domains, including **google.com** at IP **142.250.184.238**, **settings-win.data.microsoft.com** at IP **51.124.78.146**, and **watson.events.data.microsoft.com** at IP **20.42.73.29**, indicating communication with external servers, potentially part of malicious network activity.

HTTP Requests	0	Connections	25	DNS Requests	3	Threats	0	Filter by IP or domain	PCAP
Timeshift	Status	Rep	Domain	IP					
BEFORE	Responded	✓	google.com	142.250.184.238					
8220 ms	Responded	✓	settings-win.data.microsoft.com	51.124.78.146					
18428 ms	Responded	✓	watson.events.data.microsoft.com	20.42.73.29					

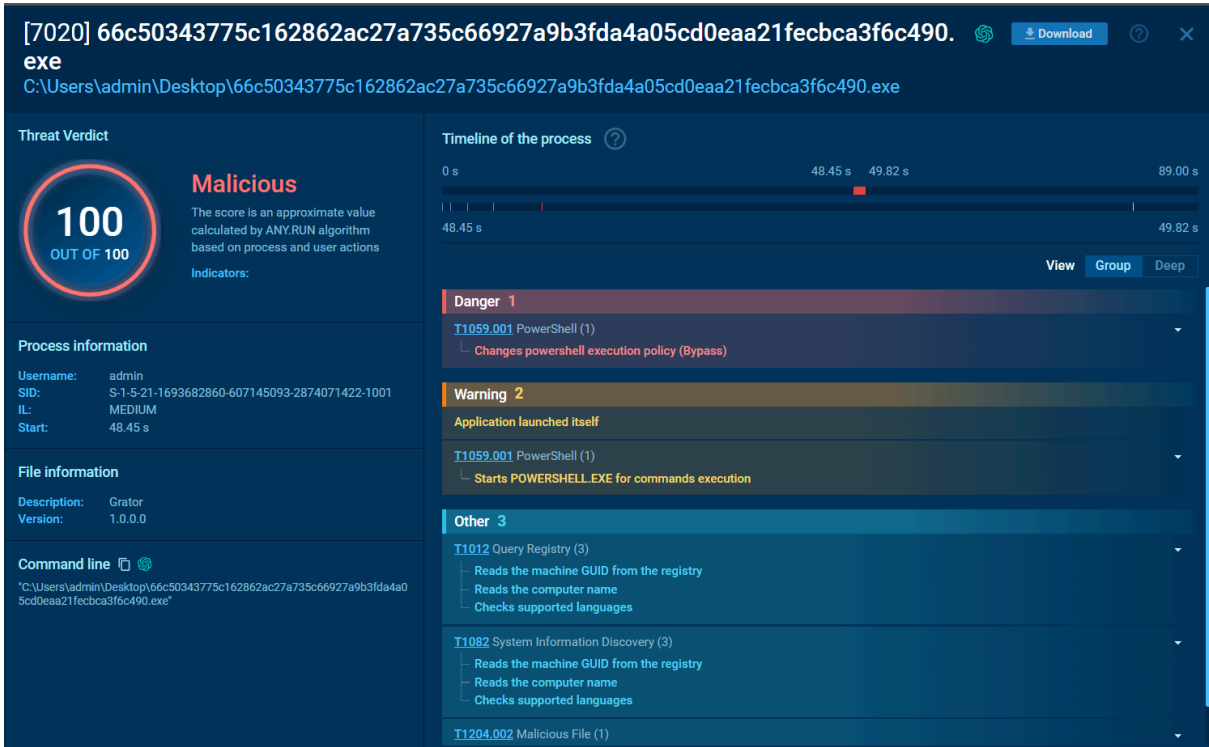
Several suspicious processes were observed, including **WinRAR.exe** and **powershell.exe**, executing commands to copy the malicious file to critical directories. The process **WerFault.exe** was also active, possibly indicating system faults triggered by the malware, suggesting persistence and further malicious activity.

Processes	Filter by PID or name	Only important
4804	WinRAR.exe C:\Users\admin\AppData\Local\Temp\66c50343775c162862ac27a735c66927a9b3fda4a05cd0eaa21fecb...	2k 1k 105
6740	66c50343775c162862ac27a735c66927a9b3fda4a05cd0eaa21fecbca3f6c490.exe PE	839 139 74
8	powershell.exe -ExecutionPolicy Bypass -command Copy-Item 'C:\Users\admin\AppData\Local\Temp\Rar\$EX...	2k 3k 101
4100	conhost.exe 0xffffffff -ForceV1	33 9 20
6320	66c50343775c162862ac27a735c66927a9b3fda4a05cd0eaa21fecbca3f6c490.exe PE	63 4 33
6840	WerFault.exe -u -p 6320 -s 548	4k 5k 181
7020	66c50343775c162862ac27a735c66927a9b3fda4a05cd0eaa21fecbca3f6c490.exe PE	834 139 74
236	powershell.exe -ExecutionPolicy Bypass -command Copy-Item 'C:\Users\admin\Desktop\66c50343775c162862...	2k 3k 102
3980	conhost.exe 0xffffffff -ForceV1	33 9 20

Upon closely looking into the processes, several suspicious activities were detected involving the executable malware. The process exhibited behavior such as modifying PowerShell execution policies, launching PowerShell commands, and querying system information. The presence of these activities, combined with the file's execution in a temporary directory, raises concerns about potential malware persistence and system compromise.



The executable on the desktop is flagged as highly malicious. It shows behavior linked to REMCOS, including screenshot capture and non-standard port connections. YARA rules confirm its malicious nature, indicating potential remote access and data exfiltration.



The following indicators of compromise (IOCs) have been identified:

Archive File:

`66c50343775c162862ac27a735c66927a9b3fda4a05cd0eaa21fecbca3f6c490.zip`

- MD5: `d0fdd7cb012019bb27a3ee604dc7ca41`
- SHA1: `ff7f3ccf129f95c3b0b1f8322d1e338d201fa1f5`
- SHA256: `78f30d2a1542078a87e751f075d96985029ea0e975c4f7696c8fbb1f99f57c0b`

Dropped Executable:

`66c50343775c162862ac27a735c66927a9b3fda4a05cd0eaa21fecbca3f6c490.exe`

- SHA256: `78f30d2a1542078a87e751f075d96985029ea0e975c4f7696c8fbb1f99f57c0b`

Malicious Connections:

- 107.175.229.139
- 20.42.73.29
- 40.127.240.158

IOCs		×
Summary of indicators of compromises 7		
☐	Copy selected	
Main object – 66c50343775c162862ac27a735c66927a9b3fda4a05cd0eaa21fecbca3f6c490.zip		^
? MD5	d0fdd7cb012019bb27a3ee604dc7ca41	
? SHA1	ff7f3ccf129f95c3b0b1f8322d1e338d201fa1f5	
? SHA256	78f30d2a1542078a87e751f075d96985029ea0e975c4f7696c8fbb1f99f57c0b	
Dropped executable file (1)		^
? SHA256	C:\Users\admin\Desktop\66c50343775c162862ac27a735c66927a9b3fda4a05cd0eaa21fecbca3f6c490.exe 66c50343775c162862ac27a735c66927a9b3fda4a05cd0eaa21fecbca3f6c490	
Connections (3)		^
IP	107.175.229.139	
? IP	20.42.73.29	
? IP	40.127.240.158	

The lifecycle of Remcos is visually represented in a graph

1. The execution starts with a user launching winrar.exe (likely a trojanized file), which triggers the process.
2. Following the execution of winrar.exe, it spawns the file 66c50343775c162862ac27a735c66927a9b3fd4a40a05cd0eaa21fecbca3f6c490.exe.

3. This executable then triggers powershell.exe, which is often used to execute malicious scripts or commands in the background.
4. Powershell.exe spawns a conhost.exe process, likely used for command-line operations without alerting the user.
5. Alongside, the executable also spawns werfault.exe, a legitimate Windows process that's being exploited, indicating an attempt to evade detection by blending in with system processes.
6. The malware ultimately leads to the execution of Remcos, a known Remote Access Trojan (RAT), which allows attackers to take control of the infected machine remotely.



The **MITRE ATT&CK Matrix** analysis of the **Remcos RAT**, highlights various tactics and techniques used during its lifecycle.

- **Execution:**
 - Command and Scripting Interpreter (PowerShell) is used for running malicious commands, noted with **4 events**.
 - User Execution (Malicious File) refers to the initial execution triggered by the victim opening a malicious file (**1 event**).
- **Persistence:**
 - Boot or Logon Autostart Execution (Registry Run Keys/Startup Folder), where the malware ensures persistence by modifying registry keys or placing files in startup locations (**1 event**).
- **Privilege Escalation:**
 - Boot or Logon Autostart Execution with Registry Run Keys is also used here, indicating overlap with persistence techniques for gaining elevated privileges.
- **Discovery:**
 - The malware performs Query Registry actions (**21 events**) and System Information Discovery (**9 events**) to gather critical system information for further operations.

- **Command and Control (C&C):**

- Non-Standard Port is used to communicate with the attacker's server, often to evade detection by traditional security measures **(2 events)**.



Tailoring the activities with the tools, we get:

1. **Process Explorer:** The malware executed from the Temp directory and used PowerShell commands to copy itself to the Startup folders, establishing persistence. The use of WerFault.exe suggests an attempt to mimic legitimate processes or handle execution errors.
2. **Regshot:** Changes tracked include the creation of multiple executables and modification of critical system files, including registry entries for startup persistence. Modified PowerShell scripts and crash dump files suggest the malware's attempts to evade detection and maintain functionality.
3. **Process Monitor:** Real-time monitoring showed that the malware spawned multiple processes and used PowerShell commands extensively to copy files into critical system directories, indicating a sophisticated persistence mechanism. File system modifications and encryption routines demonstrate its destructive intent.
4. **FakeNet-NG:** Network analysis revealed encrypted communication with a remote C2 server over port 443, linking the destination IP to a known malicious domain. This indicates data exfiltration or command reception.
5. **Netstat:** Detected outbound connections to malicious IP addresses over HTTPS, reinforcing the malware's data exfiltration or command reception capabilities.
6. **Dumplt:** Memory analysis uncovered extensive in-memory operations, suggesting that some of the malware's payload evades traditional detection methods.

Hybrid Analysis: The combined static and dynamic analysis demonstrates that the malware is a well-designed threat with capabilities for persistence, data exfiltration, and system compromise. It employs advanced techniques to evade detection and maintain control over the infected environment.

6. Comparative Analysis

Reviewing Existing Reports:

Analysis reports from the ANY.RUN public submissions platform were reviewed for the malware sample. These reports provided additional context and confirmed many of the observed behaviors:

1. [Task 1](#)
2. [Task 2](#)

7. Conclusion

The analysis of the malware sample identifies it as a sophisticated .NET-based executable, employing advanced evasion techniques and persistence strategies. Key findings include its use of encrypted communication with a remote C2 server, establishment of persistence through Startup folder modifications, and reliance on in-memory operations to evade detection.

Impact:

The malware compromises system integrity, facilitates data exfiltration, and disrupts normal operations by embedding itself deeply within the system and employing stealth techniques.

Mitigation Strategies:

To address the malware, implement comprehensive monitoring to detect unusual file, process, and network activity. Isolate affected systems, quarantine the malware, and restrict unauthorized processes to contain the threat. Eradicate the malware by removing it, cleaning residual files and registry entries, and applying necessary patches. Finally, restore systems from clean backups, enhance security measures, and conduct ongoing monitoring to ensure no residual threats remain.

8. Incident Response Plan

1. Detection

Objective: Identify the presence of the malware and assess the extent of the compromise.

- **Monitor Alerts:** Use SIEM tools (e.g., Wazuh) to review alerts for suspicious activities, such as unusual file modifications, PowerShell commands, or network traffic to known malicious domains.
- **File Integrity Check:** Utilize file integrity monitoring tools to detect unauthorized changes in critical system files and directories.
- **Network Traffic Analysis:** Analyze network traffic for encrypted communication over port 443 to identify potential C2 server interactions.
- **Process Monitoring:** Use Process Explorer and similar tools to identify unusual processes and file activities, particularly those involving PowerShell and executables in Startup folders.
- **Memory Analysis:** Capture and analyze memory dumps to detect in-memory artifacts or processes associated with the malware.

2. Containment

Objective: Limit the spread and impact of the malware within the environment.

- **Isolate Affected Systems:** Disconnect compromised systems from the network to prevent further communication with the C2 server and halt data exfiltration.
- **Identify and Quarantine Malicious Files:** Use antivirus or anti-malware tools to quarantine detected malware files. Ensure that any files copied to Startup folders are removed.
- **Restrict Malicious Processes:** Terminate processes associated with the malware using Process Explorer or similar tools.
- **Disable Unauthorized Scripts:** Remove or disable any unauthorized PowerShell scripts and other files created by the malware.
- **Network Segmentation:** Implement network segmentation to limit the malware's ability to spread to other systems.

3. Eradication

Objective: Remove the malware and all traces of its presence from the environment.

- **Complete Malware Removal:** Use reputable antivirus or anti-malware tools to perform a full system scan and ensure the removal of all detected malware components.
- **Clean Up Residual Files:** Manually verify and delete any residual malicious files, including those placed in Startup folders and other critical system locations.
- **Registry and System Cleanup:** Restore any altered registry settings and system configurations to their original state. Remove any registry entries created by the malware.
- **Update and Patch Systems:** Ensure that all systems are updated with the latest security patches to close any vulnerabilities exploited by the malware.
- **Verify System Integrity:** Perform integrity checks on the affected systems to ensure that no residual malware components remain.

4. Recovery

Objective: Restore normal operations and ensure the system is secure from future incidents.

- **Restore from Backup:** If necessary, restore affected systems from clean backups taken before the incident. Verify the integrity of backups before restoration.
- **Monitor Systems:** Implement enhanced monitoring on affected systems to detect any signs of residual malware or new threats. Use SIEM tools to continuously monitor for unusual activities.
- **Review and Strengthen Security Controls:** Review existing security controls and update them based on lessons learned from the incident. Implement additional measures, such as enhanced file integrity monitoring and network traffic analysis.
- **Conduct a Post-Incident Review:** Perform a thorough post-incident review to assess the effectiveness of the response and identify areas for improvement. Document lessons learned and update incident response plans accordingly.
- **Communicate with Stakeholders:** Inform relevant stakeholders, including management and affected parties, about the incident, its impact, and the steps taken to resolve it.

Additional Considerations:

- **Incident Documentation:** Maintain detailed records of all actions taken during the incident response, including detection methods, containment actions, eradication steps, and recovery processes.

- **Legal and Compliance:** Ensure compliance with relevant legal and regulatory requirements, including reporting obligations and data breach notifications.

LinkedIn

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